

1 Inputs, all cached on disk



2 Build the gold: which regions does each node occupy?

automatic proposal

1. locate each AST node's text inside the stream of PDF words
2. every word it covers takes that node as its owner
3. every region collects the words whose centres fall inside it
4. a region joins node n only if n owns **more than half** of that region's owned words

GOLD, for every gradeable node:

$\text{node} \rightarrow \{\text{the regions it occupies}\}$ or ABSENT, meaning it is on no page
each tagged with a **tier** — *human* > *audited* > *oracle* (unreviewed) — and a **kind**
prose / heading / formula / table / image / marker. Plus the set of **furniture** regions that belong to no node at all.

Never scored, because no trustworthy gold exists and silence is not evidence: a node the proposal could not place, one whose proposal you rejected, one that dominates no region, and unclaimed regions labelled table or formula, where the word-level proposal is structurally blind.

human verdicts

excluded:<node>
these regions | it appears nowhere

audit:<node>
accept the proposal | reject it

null:<region>
it is furniture | it belongs to node n

a human verdict always
overrides the proposal

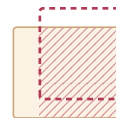
3 What a method predicts

run one aligner, *positional* or *order-free*,
on the same AST and the same regions

it returns a **reverse map**:
 $\text{region} \rightarrow \{\text{owner nodes, best first}\}$

keep only each region's **primary** owner, as
the viewer colours it, and invert:
 $\text{node} \rightarrow \{\text{predicted regions}\}$

then compare *areas*, not lists of boxes:



$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|}$$

localised when ≥ 0.5

Gold G solid, prediction P dashed. Areas are compared so a node is not punished when the detector splits its text into a different number of regions than the gold does.

4 Reported per document, then pooled over the corpus

localisation accuracy — the share of nodes reaching $\text{IoU} \geq 0.5$ · **mean IoU** · **exact** — identical region sets · **missed** — gold exists, nothing predicted

absent accuracy — correctly predicted nothing · **false-claim rate** — furniture regions some node claimed anyway · every figure also split **by node kind** and **by tier**